

Joint Project Journal (for the Programming deliverable only)

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Using the template below, make an entry for each day of the Joint Project. If you did nothing for the Programming deliverable, leave that day blank.

Link for screen cast: [Recording-20260423_154957.webm](https://www.youtube.com/watch?v=Recording-20260423_154957.webm)

Tuesday 24th March

Approximate time spent (hrs): 1:30

1. Goals for this session

Completed Kens first deliverable

2. Reflection on work completed:

3. Problems encountered

Link to final github commit for this session:

<Insert here the URL of your most recent github commit>

Once you completed your initial design and started coding, you should make multiple daily commits.

Wednesday 25th March

Approximate time spent (hrs): 1:00

1. Goals for this session

Add the template for the base game and drawing a basic shape on the screen to act as the thief until get the sprites

2. Reflection on work completed:

I was able to make a new class for the player and I called I Thief since that is what the role of the player is I added some of the functions I needed such as update render initSprites and then created a simple rectangle shape in the initSprites function this let me choose the position, color and size of the rectangle and then I drew it in the render function and called render in the update function then in the Game class I got the object of Thief and was easily able to call the update function and render the rectangle shape.

3. Problems encountered

The rectangle wasn't rendering onscreen I thought it would work since I rendered and drew it in the Thief class but I realised I had to render it in the Game class

Link to final github commit for this session:

[added project base and a simple rectangle · ITCGamesProg2/jp26-laurenconor@8c050db](#)

Wednesday 26th March

Approximate time spent (hrs): 1:00

1. Goals for this session

Get movement working for the thief

2. Reflection on work completed:

In this session I added the movement for the thief I started by getting each WASD key press so I can figure out which way the player wants to move. I then added the speed of the thief that makes it move based on the desired direction, I can edit the speed if I think it would make the game better but for now its just at 1 this is also a good speed because then the movement will be smooth

3. Problems encountered

The rectangle wasn't rendering onscreen I thought it would work since I rendered and drew it in the Thief class but I realised I had to render it in the Game class once I fixed that I was able to move the rectangle properly.

Link to final github commit for this session:

Once you completed your initial design and started coding, you should make multiple daily commits.

Wednesday 27th March

Approximate time spent (hrs): 1:00

1. Goals for this session

Add the guard class, sprite and finished off the background sprite

2. Reflection on work completed:

In this session I added a class for the guard, I used most of what I had in the thief class, except I changed the movement function. I started by using a switch statement and Enums to get the direction of the thief then the guard moves in a pre defined way, I was going to make the guard move in random patterns but then I figured it would be better for them to move left to right or up to down. I made another function so I could just move the first guard left and right, i will add more guards in this direction but I'm just starting with one at the moment to make sure every feature is working the way I want it to.

3. Problems encountered

The way I was trying to move the guard wasn't working I tried the `m_speed +=1`; but it gave me errors so i ended up with

```
case GuardDirection::DOWN:
```

```
m_guard.setPosition(sf::Vector2f{ m_guard.getPosition().x ,m_guard.getPosition().y + 1 });  
//moves guard down
```

```
break;
```

I set the position to a x and y position, for the x movement I keep the y the same and then for the y movement I keep the x the same it is simple but it definitely wasn't my first option, I think its actually cool the way there is so many different ways to code the same thing.

Link to final github commit for this session:

Wednesday 27th March

Approximate time spent (hrs): 4.20

1. Goals for this session

Implement the scent trail and hopefully the cone.

2. Reflection on work completed:

In this session, I eventually managed to get the scent trail completed for the thief, I also attempted to create the thief sprite. I'll have to work on those a bit. I started the scent trail by making a new class and making a function called `initParticles()`; this is where I set the fill colour and radius, I didn't set the position yet because I was going to do it in the next function which was `setParticlePos()`; I added the variables I thought I would need such as the `sf::CircleShape` and set that to an array like `sf::CircleShape m_particles[TOTAL_PARTICLES]`; I set the `TOTAL_PARTICLES` to 20 at first because I thought they would be small enough on the screen since I set the radius to 10, turns out they are a decent size. I might change the size of them later but it was much easier to see them while I was working on the scent trail.

I then looped through each circle shape and set the position to the `thiefPos`, I did this by passing the position of the thief to the `scentTrail` function I also passed the double `dt` which helped with the particles "expiring". I got the lifetime of each particle/ bubble and the total lifetime is around 4 seconds, over the span of those 4 seconds the transparency increases in the bubbles and once they are completely invisible they respawn under the player and it gives the scent trail effect.

The part that took me the longest was probably getting the time and figuring out how to reduce the alpha transparency on the colour, I had to use google to get my head around it and after a few tries I eventually got it. I got the fill colour of the circle in the array and set it to `sf::Color col`, then I used `static_cast<unsigned char>` which got the alpha and then multiplied it by the time left. Then I set the circle to that colour.

I think I will work on the cone implementation in the next session.

3. Problems encountered

I wanted to find the most efficient way to complete it so I spent a while going through each section and attempting to simplify it.

I also had the circles moving in the direction of the thief, which made them move when I just wanted them to spawn, stay in place and disappeared so I got rid of it and it and the movement was smooth.

Link to final github commit for this session:

[added scent trail to thief · ITCGamesProg2/jp26-laurenconor@3363ff7](#)

Wednesday 1st April

Approximate time spent (hrs): 0:30

1. Goals for this session

Add the class for vision cone and try to get it drawing

2. Reflection on work completed:

Added the new class called VisionCone, then added the cone shape set the size and colour, instead of using sf::Color::Red, I just put RGBA values so I can see which alpha value works the best for my game since I'm going to be using a scent trail.

3. Problems encountered

I was unsure on how to draw a triangle since there was no pre-set shapes in SFML for a triangle so after doing a quick search on the SFML website I learned the easiest way was to get a sf::CircleShape which is a way to set polygons and then I set the number of sides to three for the triangle and the length to 80.

Link to final github commit for this session:[started vision cone · ITCGamesProg2/jp26-laurenconor@a7d9214](#)

Friday 3rd April

Approximate time spent (hrs): 0:30

1. Goals for this session

Fixed the vision cone

2. Reflection on work completed:

Changed the vision cone from a circle shape to a convex shape then set the points which was difficult I just changed some of the points slowly until it looked somewhat like a flashlight. I changed the flashlight to a yellow colour

3. Problems encountered

When I tried to find a yellow colour I used a colour picker and found a nice shade then I put 355 by accident I figured it out after getting an error code.

Link to final github commit for this session:[added proper vision cone- with triangle fan · ITCGamesProg2/jp26-laurenconor@b613aaf](https://github.com/ITCGamesProg2/jp26-laurenconor@b613aaf)

Friday 3rd April

Approximate time spent (hrs): 1:30

1. Goals for this session

Change the vision cone based on direction guard is moving

2. Reflection on work completed:

I added a bool to check if the guard was moving left or right and returned it to the visionCone class in there I added a function to update the direction of the cone I added a if statement to get the cone for whichever direction the guard is facing, in initCone I set the point counter to 5 for each point and updated the cone direction there.

In the constructor I needed to get a copy of the guard so I could get the position and place the cone in the center so it looks like a flashlight.

I created a second guard and got it to move up and down, I had to change the constructor to initialise the starting direction to LEFT and initialised the starting position for each of the guards

3. Problems encountered

I struggled getting a flipped version of the cone I had already had I thought it work if I made the numbers on the left and right negative but it was just the numbers on the left I had to change.

Got confused with the bool, even if the player was moving right the flashlight was pointing left.,

Link to final github commit for this session:[added second guard · ITCGamesProg2/jp26-laurenconor@376f9c8](https://github.com/ITCGamesProg2/jp26-laurenconor@376f9c8)

Saturday 4th April

Approximate time spent (hrs): 2:30

1. Goals for this session

Add collision checking between cone and scent trail.

2. Reflection on work completed:

Firstly in the scent trail class I made a bool function called intersects and that ran through the array of circles, I did some error checking to see if the colour was completely faded on each circle, which meant if the circle was invisible it wouldn't interfere with the boundary checking. I then checked to see if the trail intersected with the global bounds of the cone and return true if it did.

In the vision cone class I returned I made another bool class and checked if they intersected and put it in the update function, if they did intersect I made a new function called handle collision which changes the colour.

I made a Enum for each state of the vision cone, searching, attacking, persuing and changed the colours if they were true.

3. Problems encountered

Was constantly colliding with every circle even if I couldn't see it anymore so I used an if statement like - if (m_particles[parNum].getFillColor().a == 0) which fixed that problem.

It checks if the colour value for the alpha is zero which means the circle is transparent/ invisible.

Link to final github commit for this session:[added collision checking and state mechanism · ITCGamesProg2/jp26-laurenconor@2dbf040](https://github.com/ITCGamesProg2/jp26-laurenconor@2dbf040)

Thursday 9th April

Approximate time spent (hrs): 2:00

1. Goals for this session

Was working on Kens part of project2.

Reflection on work completed:

3. Problems encountered

Link to final github commit for this session:

Tuesday 14th April

Approximate time spent (hrs): 2:00

1. Goals for this session

Fix the vision cone

2. Reflection on work completed:

I realised I used the wrong thing for the vision cone so I had to fix it. In vision cone I changed the .setPoint to a startAngle and set the cone as a sf::PrimitiveType::TriangleFan, I resized it and set the position. I got the angle of the cone using float angle and that went through the loop for each point and I also set the corner of the cones position as the guard position so it looks like a flashlight.

I fixed how I set the colours for each point. Instead of

```
m_cone.setFillColor({ 102,255,102,155 });
```

I used m_colour = { 102,255,102,155 };

m_color sets the colour for each point on the cone.

I wanted to work on the cones states more so in the guard class I made two new functions one to check if anything intersected with it and a return bounds function this will make it easy to check in the vision cone class. I already had a checking for trail collision that checks if the vision cone intersected with the alive trail bubbles that switches the state to ALERT

If the trail intersects with the guards bounds the state is changed to PERSUING I need to add a perusing function where the guard follows the player and finally if the guard intersects with the thief the state is ATTACKING

3. Problems encountered:

I had an issue where my game wasn't running due to a .dll error so I asked for help and had to grab files from one of my last commits and put it in the debug folder. That fixed the main problem. The last problem i had was right after I fixed the problem I had an error where I was

doing a for loop and had a 9 instead of 'i' for the index, I was able to check everything was running again

Link to final github commit for this session:

[fixed vision cone · ITCGamesProg2/jp26-laurenconor@eeb7734](#)

[vision cone fixed and added collision checking · ITCGamesProg2/jp26-laurenconor@6986715](#)

Wednesday 15th April

Approximate time spent (hrs): 1:30

1. Goals for this session

Add an items class so I can create the collectible items for the thief to steal.

2. Reflection on work completed:

I created a new class called items and created functions such as initSprites, setPosition, getPosition and made a new ShapeRectangle, I added the class to game and passed the rectangle to it. I did that by adding the position when I added the object to the class like

```
Items m_itemOne{ sf::Vector2f{ 650,175 } };
```

And in the Items class I had to pass the position from the game class I put it in the constructor

```
Items(sf::Vector2f t_position);
```

Then I passed it to the function I wanted to use it in which was setPosition()

3. Problems encountered:

Efficiently passing the different boxes to the game class. I was getting confused why my boxes were in the exact same spot even though I positioned them in different spots, I had to pass the position since I had three boxes, I had to look through some of the Guard class to find out the best way to pass it in the item constructor I had sf::Vector2f position; and then i just had to have a starting position for them in the game class and then they were finally being drawn in different spots on the screen

Link to final github commit for this session:[added bound checking and items class · ITCGamesProg2/jp26-laurenconor@c6b1930](https://github.com/ITCGamesProg2/jp26-laurenconor@c6b1930)

Thursday 16th April

Approximate time spent (hrs): 1:00

1. Goals for this session

Get the guard to move in players direction when scent trail and cone collide

2. Reflection on work completed:

In the Guard class I added two new functions, void setChasing(bool chasing); which checked if the guard/ vision cone was intersecting with the thief's scent trail and a chaseTarget function, in the chaseTarget function, i got the direction of the guard and minused the target pos (theif position) to get the direction the guard should move in. I got the length the guard needed to go by using direction and then if the length was greater than 0 I normalised it which makes the movement smoother, then I set the position of the guard to the new position and I set the chase speed to be faster than the player but I will change it because its too hard to play it when you get caught

I created a few items and I will add them to the game soon

3. Problems encountered:

I forgot to pass the bool into m_guard.setChasing(); and the thief position into the m_guard.update in the game class so I had to fix them it was easy enough but I should have checked to make sure they were passing them.

Link to final github commit for this session:[enemy following thief on collision · ITCGamesProg2/jp26-laurenconor@313862c](https://github.com/ITCGamesProg2/jp26-laurenconor@313862c)

Thursday 16th April

Approximate time spent (hrs): 0:50

1. Goals for this session

Fix the direction the vision cone faces

Reflection on work completed:

After the last commit I noticed the vision cone wasn't completed since the up down movement was not being used properly. I changed the if statement to a switch statement and used the enum from the Guard class to change the directions, I had to test the angles for each direction a few times to make sure it worked properly

3. Problems encountered:

For the down movement I had the degrees to be 360 but it was actually 90 it was a silly mistake since 360 would almost be the same direction as 0. I also missed passing the direction through a few functions so I had to go searching for them and pass it through them, the colour was broken but I added `m_cone[0].color = m_color;` and that got the colour for each point on the cone so it fixed the issue.

Link to final github commit for this session: [fixed vision cones direction · ITCGamesProg2/jp26-laurenconor@83ae243](https://github.com/laurenconor@83ae243/ITCGamesProg2/jp26-fixed_vision_cones_direction)

Friday 17th April

Approximate time spent (hrs): 1:20

1. Goals for this session

Add sprites

2. Reflection on work completed:

In this session I had a bit of a struggle adding the sprite for the collectable items, so firstly I added `sf::Texture` and `sf::Sprite` in `Items.h`, then I went into `Items.cpp` and in the `initSprites` function attempted to load the file from the images folder like `!_jewlText.loadFromFile("ResourceFiles/Images/jewleryOne.png")`.

I set the position and set the texture to the sprite, I got errors telling me to add `sf::Sprite` to the constructor and I did that and it told me to add it to the game constructor.

After many failed attempts of going around and trying to fix it I eventually got it by setting the texture to the sprite in the header file and adding the file path in the constructor. I drew the sprite on top of the boxes so it looks like they are in a display cabinet. I just need to make a few more

sprites so it looks nicer my overall goal is to have three different items in each display case so the thief can steal them to get more points.

3. Problems encountered:

Drawing the sprites had to use the constructor to get the texture which took a while to fix.

Adding the third guard, I created the object in the Game.h file and then in the Game.cpp I forgot to update it and add the cone so it was just the red rectangle spawning and doing nothing for a minute.

Link to final github commit for this session:[added new sprites to game & new guard · ITCGamesProg2/jp26-laurenconor@4594d19](https://github.com/ITCGamesProg2/jp26-laurenconor@4594d19)

Saturday 18th April

Approximate time spent (hrs): 3:00

1. Goals for this session

Fix the movement in the vision cone to follow the player when attacking/pursuing

2.Reflection on work completed:

I changed the origin of the vision cone to start in the center of the guard's rectangle

`sf::Vector2f origin = { guardPos.x + 15.0f, guardPos.y + 35.0f };` since before it was a little off center. I moved the code I used to get the left right up and down seperate cones and put it into a new function called `directionToAngle` and passed the `GuardDirection` state to it i used a switch statement and returned the direction.

In `initCone` I passed the `guardDirection` and called the `updateConeDirection(guardPos, directionToAngle(dir));` . In the update function I have an if statement checking if the state `SEARCHING` or `ALERT` were true and it updates the cone to face the player.

I had to get the `startingAngle` in the `checkCollision` function, firstly I got the difference in the thief position and guards position, the `angleRad` got the arc of them and that helped to get the angle degrees by multiplying it by $(180.0f / \text{PI})$; then I minused the angle degree by $(\text{fieldOfView} / 2.0f)$

In the guard.cpp I checked if the direction the guard was facing was correct in the chaseTarget function by getting the absolute value of the direction.x and if it was less than the absolute value of direction.y like

```
if (std::abs(direction.x) > std::abs(direction.y))
```

And if it was less than 0 the direction would be RIGHT else LEFT and if the dir.x > dir.y was false it checked the dir.y for UP and DOWN. In the if statement in LRMovement function I added m_guard.setPosition({ 0,m_guard.getPosition().y }); to make sure it was moving left and m_guard.setPosition({ 1410,m_guard.getPosition().y }); if it was moving right

3. Problems encountered:

There was a certain point I had the cone facing the wrong direction entirely when it was following the thief, it was because I needed to add an if statemet to move the cone in the direction of the thief using the directionToAngle(dir).

I realised there was a fault where a guard moving right and left started following me, as I was moving down the guard lost track of my scent and then kept moving down past the screen so I added error checking. If the guard went hit the bounds it changed direction to the opposite direction, I made a new Enum called GuardMovement {LEFTRIGHT, UPDOWN} and passed the basic movement in the guard constructor. In game I set the GuardMovement to LEFTRIGHT or UPDOWN and now if they follow me and change their original direction they won't go past the screen and they will be able to move in the new state direction

Link to final github commit for this session:[vision cone follows player when persuing · ITCGamesProg2/jp26-laurenconor@5293f0b](https://github.com/ITCGamesProg2/jp26-laurenconor@5293f0b)

Sunday 19th April

Approximate time spent (hrs): 1:00

1. Goals for this session

Get the collision for the thief and boxes

2.Reflection on work completed:

In the Items class I returned the global bounds of the box, made a new function called wallChecking, it goes through each box and it checks if the thief is colliding, if its not then it continues and if it is then it returns true. In init in the game class we check to see if we are colliding with any box

```
m_thief.setBox(&m_itemOne);  
m_thief.setBox(&m_itemTwo);  
m_thief.setBox(&m_itemThree);
```

I then needed to stop the thief from going into the boxes so I made a new variable sf::Vector2f pos = m_thief.getPosition(); and replaced m_thief.getPosition,x /y with pos.x/y and if it was colliding, set the position to the position it was currently in so it wouldn't run though the wall.

I also added a few new sprites and drew them on the boxes.

3. Problems encountered:

```
if (wallChecking())  
{  
  
m_thief.setPosition(m_thief.getPosition());  
}
```

I thought this would work but it was much easier to just use pos

Link to final github commit for this session:[added box collision checking · ITCGamesProg2/jp26-laurenconor@f10aaba](https://github.com/ITCGamesProg2/jp26-laurenconor@f10aaba)

Sunday 19th April

Approximate time spent (hrs): 1:00

1. Goals for this session

Get the thief's lives. The guard's can kill the thief

2.Reflection on work completed:

I added new variables ints for the total lives for the timer and a bool for if the player is alive. I made a bool function to return if the thief was alive. Then I made another function to take away

the lives using a timer, in vision cone, if the guard is attacking the thief a life goes down every seconds and the loseLife function resets every seconds, i then returned the lives so I could draw them on the screen, I added the sf::Text to game.h and set the font in the header file, I then got the position, size and drew it

3. Problems encountered:

I forgot to add the timer so the thief lost the lives as soon as they hit the guard

Link to final github commit for this session:[added thieves lives and death · ITCGamesProg2/jp26-laurenconor@4d54c61](https://github.com/laurenconor@4d54c61/ITCGamesProg2/jp26-added-thiefs-lives-and-death)

Tuesday 21st April

Approximate time spent (hrs): 1:00

1. Goals for this session

Adding the sprites for the guard's and thief.

2. Reflection on work completed:

Firstly I had to add sf::Sprite and sf::Texture to both the Guard and Thief classes, in init sprites I did some error checking with the sprites to check if they were loading properly or not

```
if (!m_guardTexture.loadFromFile("ResourceFiles/Images/walking_guard.png"))  
{  
    std::cout << "Guard not loading :( " <<std::endl;  
}
```

I then changed all the m_guard to m_guardSprite. It took me a while to actually get the guards to draw properly

3. Problems encountered:

It took me a while to actually get one of the guards from the sprite sheets because I was creating the textureRect before I actually set the sprite.

Getting the proper size for the intRect, the problem was I didn't know how to get the textureRect I thought it was `m_guardSprite.setTextureRect({ 0,0,30,70 })` but I actually needed to do `m_guardSprite.setTextureRect(sf::IntRect{ { 0,0}, {60, 70} })`;

Link to final github commit for this session: [drew guard sprite sheet on screen · ITCGamesProg2/jp26-laurenconor@6116ba2](https://github.com/ITCGamesProg2/jp26-laurenconor@6116ba2)

Tuesday 21st April

Approximate time spent (hrs): 3:00

1. Goals for this session

Getting the collision between the thief and the items the player needs to steal

2. Reflection on work completed:

There is a lot of repeated code in here such as returning the individual items global bounds, setting the bool if they get picked up by the thief, getting the bool to see if they are alive/ on screen, I added the thief sprite and had to change the textureRect to be tighter to the sprite.

I needed to scan through the items in the thief class so I looped through the items in `m_boxes`

For (`Items* item :m_boxes`), if the item was empty it continued through the items if (`item == nullptr`) continue;

3. Problems encountered:

A issue I had was the sprite had a black background, I tried to remove it but it didn't erase, I needed to make a new file and removed the background that way.

Link to final github commit for this session: [item collision checking · ITCGamesProg2/jp26-laurenconor@799a3eb](https://github.com/ITCGamesProg2/jp26-laurenconor@799a3eb)

Wednesday 22nd April

Approximate time spent (hrs): 1:00

1. Goals for this session

Fixing the guard sprite, vision cone and scent trail

2. Reflection on work completed:

I changed the textureRect for the guard and fixed the sprite then I changed the origin position on the Guards vision cone to start on the guards flashlight and made the scent trail follow the thief from the center instead of the top left corner

3. Problems encountered:

Had to find the right spot for the guards vision cone to start from took me while but I got it

Link to final github commit for this session: [added new sprite and fixec cones position and scent trail position · ITCGamesProg2/jp26-laurenconor@c8ed073](https://github.com/ITCGamesProg2/jp26-laurenconor@c8ed073)

Wednesday 23rd April

Approximate time spent (hrs): 1:00

1. Goals for this session

Add messages for end of game

2. Reflection on work completed:

I added two new sprites, changed the spear image to an exit sign added a message if the player reached the exit sign and got collision checking with the thief

```
if (m_items->getCloakVisible())
```

```
{
```

```
item->setCloakVisible(false);//returns false to stop drawing the item in the items class
```

```
m_points += 5;
```

```
} and the images and the thief can “collect them” by hovering over them and the item will move off screen and the player will get points for the item
```

If the player was able to collect items and reach the exit without getting caught they would win the game and their points would convert to money at the end

3. Problems encountered:

Took me a while to figure out how to give the player points when they first collect the item instead of the whole time they were hovering I just did

```
if(m_item->getItemVisible()  
  
{m_points+5; and set the itemsVisible to false  
  
}
```

Link to final github commit for this session: [new messages, new sprites and end game · ITCGamesProg2/jp26-laurenconor@7b5b795](https://github.com/ITCGamesProg2/jp26-laurenconor@7b5b795)

Wednesday 23rd April

Approximate time spent (hrs): 1:00

1. Goals for this session

Add comments

2. Reflection on work completed:

Cleaned up everything and added comments

Link to final github commit for this session: [added more comments · ITCGamesProg2/jp26-laurenconor@a4ed9ae](https://github.com/ITCGamesProg2/jp26-laurenconor@a4ed9ae)